Kaabil Sheet - DSA Beginner Friendly

To start off - 10 Basic Array Problems

1. Find the Sum of All Elements

Given an array, print the **sum** of all elements.

2. Find the Maximum Element

Print the **largest** number in the array.

3. Find the Minimum Element

Print the **smallest** number in the array.

4. Count Even and Odd Elements

Count how many numbers are **even** and **odd**.

5. Search an Element in an Array

Given an array and a value, check if it **exists** in the array.

6. Reverse the Array

Reverse the array using a loop.

7. Find the Average of Elements

Calculate and print the **average** of array elements.

8. Count the Frequency of Each Element

Print the number of occurrences of each element.

9. Copy Array Elements to Another Array

Create a new array and copy all elements.

10. Check if Array is Sorted (Ascending)

Return **true** if sorted, otherwise false.

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10 Basic String Problems

1. Find Length of a String (Without Using builtin)

Count characters manually using loops.

2. Count Vowels and Consonants

Given a string → count **A,E,I,O,U** and consonants.

3. Convert String to Uppercase / Lowercase

Manually convert characters.

4. Reverse a String

Reverse using a loop or two pointers.

5. Check Palindrome String

Input → "madam", output → palindrome.

6. Count Words in a Sentence

Count number of words separated by spaces.

7. Remove All Spaces

Input: `"a b c"` → Output: `"abc"`.

8. Find Frequency of Each Character

Count occurrences of each character.

9. Compare Two Strings (Without builtin)

Check if both strings are equal by comparing characters.

10. Find the Largest Word in a Sentence

Input: `"I love programming"` → Output: `"programming"`.

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100 DSA problems

1) ARRAYS – 20 Problems

1. Two Sum – https://leetcode.com/problems/two-sum/
2. Best Time to Buy and Sell Stock – https://leetcode.com/problems/best-time-to-buy-and-sell-stock/
3. Contains Duplicate – https://leetcode.com/problems/contains-duplicate/
4. Maximum Subarray (Kadane's) – https://leetcode.com/problems/maximum-subarray/
5. Move Zeroes – https://leetcode.com/problems/move-zeroes/
6. Rotate Array – https://leetcode.com/problems/rotate-array/
7. Remove Duplicates from Sorted Array – https://leetcode.com/problems/remove-duplicates-from-sorted-array/
8. Merge Sorted Array – https://leetcode.com/problems/merge-sorted-array/
9. Intersection of Two Arrays II – https://leetcode.com/problems/intersection-of-two-arrays-ii/
10. Plus One – https://leetcode.com/problems/plus-one/
11. Missing Number – https://leetcode.com/problems/missing-number/
12. Majority Element – https://leetcode.com/problems/majority-element/
13. Valid Mountain Array – https://leetcode.com/problems/valid-mountain-array/
14. Squares of a Sorted Array – https://leetcode.com/problems/squares-of-a-sorted-array/
15. Reverse Array (GFG) – https://www.geeksforgeeks.org/write-a-program-to-reverse-an-array-or-string/
16. Leaders in an Array (GFG) – https://www.geeksforgeeks.org/leaders-in-an-array/

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17. Second Largest Element (GFG) – https://www.geeksforgeeks.org/find-second-largest-element-array/

18. Check if Array Sorted – https://www.geeksforgeeks.org/check-if-an-array-is-sorted/

19. Find the Frequency of an Element – https://www.geeksforgeeks.org/count-number-of-occurrences-in-a-sorted-array/

20. Array Sum (HackerRank) – https://www.hackerrank.com/challenges/simple-array-sum/problem

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2) STRINGS – 12 Problems

21. Valid Anagram – https://leetcode.com/problems/valid-anagram/

22. Valid Palindrome – https://leetcode.com/problems/valid-palindrome/

23. Reverse String – https://leetcode.com/problems/reverse-string/

24. First Unique Character – https://leetcode.com/problems/first-unique-character-in-a-string/

25. Ransom Note – https://leetcode.com/problems/ransom-note/

26. Longest Common Prefix – https://leetcode.com/problems/longest-common-prefix/

27. Implement strStr() – https://leetcode.com/problems/implement-strstr/

28. Remove Vowels (GFG) – https://www.geeksforgeeks.org/remove-vowels-from-a-string/

29. Pangram Checker (GFG) – https://www.geeksforgeeks.org/check-if-the-given-string-is-pangram/

30. Print Duplicate Characters – https://www.geeksforgeeks.org/print-all-the-duplicates-in-the-input-string/

31. String Reverse (GFG) – https://www.geeksforgeeks.org/reverse-a-string-in-python/

32. Strong Password (HackerRank) – https://www.hackerrank.com/challenges/strong-password/problem

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3) HASHING / MAPS – 6 Problems

33. Two Sum (map approach) – https://leetcode.com/problems/two-sum/

34. Group Anagrams – https://leetcode.com/problems/group-anagrams/

35. Contains Duplicate – https://leetcode.com/problems/contains-duplicate/

36. Happy Number – https://leetcode.com/problems/happy-number/

37. Word Pattern – https://leetcode.com/problems/word-pattern/

38. Count Frequencies (GFG) – https://www.geeksforgeeks.org/counting-frequencies-of-array-elements/

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4) MATH – 8 Problems

39. FizzBuzz – https://leetcode.com/problems/fizz-buzz/

40. Count Primes – https://leetcode.com/problems/count-primes/

41. Power of Three – https://leetcode.com/problems/power-of-three/

42. Add Digits – https://leetcode.com/problems/add-digits/

43. Palindrome Number – https://leetcode.com/problems/palindrome-number/

44. Reverse Integer – https://leetcode.com/problems/reverse-integer/

45. GCD of Two Numbers (GFG) – https://www.geeksforgeeks.org/c-program-find-gcd-two-numbers/

46. LCM of Two Numbers (GFG) – https://www.geeksforgeeks.org/program-to-find-lcm-of-two-numbers/

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5) TWO POINTERS – 6 Problems

47. Two Sum II – https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/

48. Reverse String – https://leetcode.com/problems/reverse-string/

49. Valid Palindrome II – https://leetcode.com/problems/valid-palindrome-ii/

50. Container With Most Water – https://leetcode.com/problems/container-with-most-water/

51. Remove Element – https://leetcode.com/problems/remove-element/

52. Sort Colors (Dutch Flag) – https://leetcode.com/problems/sort-colors/

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6) SLIDING WINDOW – 6 Problems

53. Maximum Average Subarray I – https://leetcode.com/problems/maximum-average-subarray-i/

54. Max Consecutive Ones – https://leetcode.com/problems/max-consecutive-ones/

55. Longest Substring Without Repeat – https://leetcode.com/problems/longest-substring-without-repeating-characters/

56. Minimum Size Subarray Sum – https://leetcode.com/problems/minimum-size-subarray-sum/

57. Sliding Window Maximum – https://leetcode.com/problems/sliding-window-maximum/

58. Longest Repeating Character Replacement – https://leetcode.com/problems/longest-repeating-character-replacement/

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7) STACK – 5 Problems

59. Valid Parentheses – https://leetcode.com/problems/valid-parentheses/

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60. Min Stack – https://leetcode.com/problems/min-stack/

61. Next Greater Element I – https://leetcode.com/problems/next-greater-element-i/

62. Implement Stack Using Queues – https://leetcode.com/problems/implement-stack-using-queues/

63. Balanced Parentheses (GFG) – https://www.geeksforgeeks.org/check-for-balanced-parentheses-in-an-expression/

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8) QUEUE – 4 Problems

64. Implement Queue Using Stacks – https://leetcode.com/problems/implement-queue-using-stacks/

65. Circular Queue – https://leetcode.com/problems/design-circular-queue/

66. Queue Using Array (GFG) – https://www.geeksforgeeks.org/array-implementation-of-queue-simple/

67. Rotten Oranges – https://leetcode.com/problems/rotting-oranges/

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9) LINKED LIST – 10 Problems

68. Reverse Linked List – https://leetcode.com/problems/reverse-linked-list/

69. Middle of Linked List – https://leetcode.com/problems/middle-of-the-linked-list/

70. Merge Two Sorted Lists – https://leetcode.com/problems/merge-two-sorted-lists/

71. Delete Node in LL – https://leetcode.com/problems/delete-node-in-a-linked-list/

72. Palindrome Linked List – https://leetcode.com/problems/palindrome-linked-list/

73. Linked List Cycle – https://leetcode.com/problems/linked-list-cycle/

74. Remove Duplicates (GFG) – https://www.geeksforgeeks.org/remove-duplicates-from-an-unsorted-linked-list/

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75. Intersection of Two Linked Lists – https://leetcode.com/problems/intersection-of-two-linked-lists/

76. Add Two Numbers – https://leetcode.com/problems/add-two-numbers/

77. Detect Loop (GFG) – https://www.geeksforgeeks.org/detect-loop-in-a-linked-list/

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10) TREES – 10 Problems

78. Maximum Depth of Binary Tree – https://leetcode.com/problems/maximum-depth-of-binary-tree/

79. Same Tree – https://leetcode.com/problems/same-tree/

80. Symmetric Tree – https://leetcode.com/problems/symmetric-tree/

81. Invert Binary Tree – https://leetcode.com/problems/invert-binary-tree/

82. Binary Tree Level Order Traversal – https://leetcode.com/problems/binary-tree-level-order-traversal/

83. Path Sum – https://leetcode.com/problems/path-sum/

84. BST Search – https://leetcode.com/problems/search-in-a-binary-search-tree/

85. Insert into BST – https://leetcode.com/problems/insert-into-a-binary-search-tree/

86. Count Leaf Nodes (GFG) – https://www.geeksforgeeks.org/count-leaves-in-binary-tree/

87. Height of Binary Tree (GFG) – https://www.geeksforgeeks.org/height-of-binary-tree/

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11) SORTING – 5 Problems

88. Bubble Sort (GFG) – https://www.geeksforgeeks.org/bubble-sort/

89. Insertion Sort (GFG) – https://www.geeksforgeeks.org/insertion-sort/

90. Selection Sort (GFG) – https://www.geeksforgeeks.org/selection-sort/

91. Merge Sort (GFG) – https://www.geeksforgeeks.org/merge-sort/

92. Quick Sort (GFG) – https://www.geeksforgeeks.org/quick-sort/

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12) RECURSION – 5 Problems

93. Factorial – https://www.geeksforgeeks.org/program-for-factorial-of-a-number/

94. Fibonacci – https://leetcode.com/problems/fibonacci-number/

95. Print Subsets – https://www.geeksforgeeks.org/subsets-of-a-given-set/

96. Tower of Hanoi – https://www.geeksforgeeks.org/c-program-for-tower-of-hanoi/

97. Sum of Digits – https://www.geeksforgeeks.org/recursive-programs-to-find-the-sum-of-digits-of-a-number/

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13) BASIC DYNAMIC PROGRAMMING – 3 Problems

98. Climbing Stairs – https://leetcode.com/problems/climbing-stairs/

99. Min Cost Climbing Stairs – https://leetcode.com/problems/min-cost-climbing-stairs/

100. House Robber – https://leetcode.com/problems/house-robber/

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Resources

Free Documentation / Tutorials (Text-Based)

- GeeksforGeeks – “DSA Tutorial - Learn Data Structures and Algorithms” (free)

  https://www.geeksforgeeks.org/dsa/dsa-tutorial-learn-data-structures-and-algorithms/ (GeeksforGeeks)

- Programiz – “Learn Data Structures and Algorithms” (step-by-step free tutorial) https://www.programiz.com/dsa (Programiz)

- W3Schools – “DSA Tutorial” (free interactive examples) https://www.w3schools.com/dsa/ (W3Schools)

- TutorialsPoint – “Data Structures & Algorithms (DSA) Tutorial” https://www.tutorialspoint.com/data_structures_algorithms/index.htm (Tutorials Point)

YouTube Channels / Video Tutorials

- NeetCode (YouTube): “Learn Algorithms & Data Structures for FREE” https://www.youtube.com/c/neetcode (YouTube)

- takeUforward (YouTube) – highly recommended in the resource list for DSA. (DEV Community)

- The DesignGurus article lists several strong channels: Abdul Bari, MyCodeSchool, etc. (designgurus.io)

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